

Effectiveness of "Focus Lines" for Attentional Guidance in Virtual Reality

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Abstract

This study applied the radial "Focus lines" used in manga as an attentional guidance technique in VR space to examine the possibility of achieving both attentional guidance effects and immersion maintenance. In the experiment, subjects explored and collected randomly appearing target objects in a VR environment under a tracking scenario. We then compared two versions of focus lines (black and gray), with parameters set based on preliminary experiments, against arrows and lighting based on the number of obtained targets and subjective evaluations. The results showed that focus lines (gray) demonstrated a significantly higher number of obtained targets and the attentional guidance effects compared to no visual cue, arrows, and lighting. However, in terms of maintaining the immersion, focus lines (black, gray) received lower evaluations than no visual cue and lighting. Nevertheless, focus lines obtained positive immersion, and no significant difference was found between the more conspicuous gray and subtler black lines. This suggests that it is possible for focus lines to enhance the attentional guidance effects through parameter tuning without significantly compromising the immersion.

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1 Introduction

In recent years, VR (virtual reality) technology has been applied in a wide range of fields, including games, education, and medicine, and its popularity is growing fast. In a VR environment, users can look in any direction within a 360° field of view simply by moving their heads. As a result, objects that developers want users to focus on fall outside the field of view, and users often fail to notice them. Traditional attentional guidance include showing arrows [Wallgrün et al. 2020] in VR spaces, lighting the target object, or applying motion (e.g., shaking or pulsing) to the target object itself. However, it has been pointed out that inappropriate guidance can detract from the realism and immersion of VR spaces [McCay-Peet et al. 2012]. This study focuses on "Focus lines," commonly used in manga, and proposes a new attentional guidance for applying them

to attentional guidance in VR spaces. Focus lines extend radially from the target object, making it easy to intuitively indicate the position of the target object. It is possible to adjust the color, the number of lines, the length, the thickness, and the opacity which may reduce interference with the immersion. In this paper, we conduct comparative experiments with traditional attentional guidance (arrows, lighting) to verify the effectiveness of the attentional guidance using focus lines, and clarify their effects and impact on the VR experience.

2 Experiment

The purpose of this experiment is to quantitatively verify the guiding effects and interference with the immersion when using focus lines as an attentional guidance technique in VR space. Subjects search for target objects in VR space and acquire them. We compared the effects of the five stimulus conditions based on two indicators: the number of obtained targets, and subjective evaluations of attentional guidance and immersion.

2.1 Experimental Stimuli

Five conditions using visual stimuli were set as techniques for guiding the attention, shown in Figure 1.

- (a) **No visual cue** – A baseline condition without any guiding visual stimulus.
- (b) **Focus lines (black)** – Black radial lines displayed from the target object, with parameters set based on preliminary experiments (number of lines: 6, length: 1.6 m, thickness: 1 cm, opacity: 100%).
- (c) **Focus lines (gray)** – Gray radial lines displayed from the target object, with parameters set based on preliminary experiments (number of lines: 24, length: 0.8 m, thickness: 2 cm, opacity: 33%).
- (d) **Arrow** – An arrow indicating the direction of the target object from the user's perspective.
- (e) **Lighting** – A spotlight cast on the target object to enhance contrast against the background.

2.2 Experimental Procedure

Subjects were free to move around the VR environment while searching for target objects. To create a tense atmosphere, a bear character continuously chased the subject from behind. At the beginning of each trial, three target objects appeared simultaneously at random locations selected from nine predefined spawn points. The three target objects were assigned one of the five conditions described in Section 2.1 Each subject performed a total of fifteen trials (5 conditions × 3 trials), and the order in which the conditions were presented was randomly determined for each subject. At the end of each trial, the subjects answered a subjective evaluation

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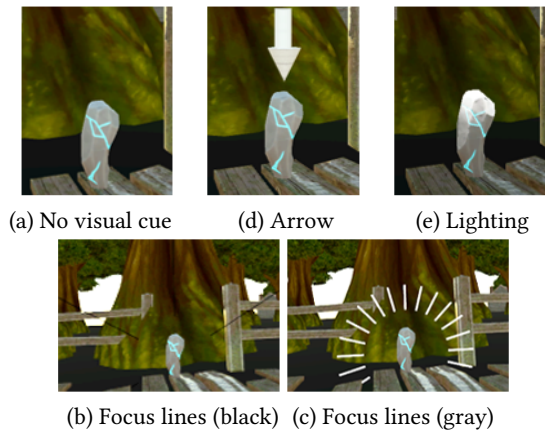


Figure 1: Experimental Stimuli (Brightness adjusted)

questionnaire regarding attentional guidance and immersion for that condition.

The questionnaire consisted of the following seven items:

- Q1: I could immediately understand what was indicated.
- Q2: It helped me find the target quickly.
- Q3: It distracted my focus.
- Q4: It drew my attention immediately.
- Q5: It interfered with the background or objects.
- Q6: I felt discomfort.
- Q7: It matched the game's atmosphere without feeling strange.

The questionnaire items consisted of three questions related to attentional guidance (Q1, Q2, and Q4) and three questions related to immersion (Q3, Q5, and Q6) selected from an existing VR experience evaluation scale [Hu et al. 2021], as well as one independently set question (Q7). Each item was evaluated on a 7-point Likert scale ranging from -3 to 3.

3 Results and Discussion

This experiment included 21 Subjects (18 males, 3 females in their 20s), and 20 were included in the analysis after one person discontinued due to VR sickness.

3.1 Number of Obtained Targets

Figure 2 (a) shows the results for the number of obtained targets. The Friedman test ($p < 0.05$) revealed significant differences between conditions. Post hoc comparisons (Wilcoxon signed-rank test, Bonferroni corrected, $p < 0.05$) showed that Focus lines (gray) were obtained significantly more often than No visual cue, Arrow, and Lighting ($p < 0.01$). Additionally, Focus lines (black) were also obtained significantly more often than Lighting. These results suggest that guidance using focus lines is effective in increasing the number of obtained targets.

3.2 Subjective Evaluation

The attentional guidance shown in Figure 2 (b) is the average score of Q1, Q2, and Q4, and the immersion maintenance shown in Figure 2 (c) is the average score of Q3, Q5, Q6, and Q7 (Q3, Q5, and Q6 are after sign reversal).

Regarding the attentional guidance, the Friedman test ($p < 0.05$) revealed significant differences between conditions. Post hoc comparisons (Wilcoxon signed-rank test, Bonferroni corrected, $p <$

0.01) showed that Focus lines (gray) received significantly higher ratings than all other four conditions. Additionally, Arrow received significantly higher ratings than Focus lines (black). Furthermore, the three conditions of Focus lines (black), Focus lines (gray) and Arrow, significantly outperformed the conditions with No visual cue and Lighting. These results suggest that Focus lines (gray) are most effective in terms of subjective attentional guidance effects.

In terms of maintaining the immersion, the Friedman test ($p < 0.05$) revealed significant differences between conditions. Post hoc comparisons (Wilcoxon signed-rank test, Bonferroni corrected, $p < 0.01$) showed that No visual cue and Lighting received the significantly highest ratings. Focus lines (black, gray) received lower ratings than No visual cue and Lighting, however, Focus lines (black, gray) obtained the positive scores. Additionally, even though Focus lines (gray) were more prominent than Focus lines (black), no significant difference was observed in terms of maintaining the immersion. This suggests that the focus lines themselves, as a comic-style effect, were sufficiently harmonized with the VR space's atmosphere where their prominence did not significantly make differences in the immersion.

The results of this study suggest that the attentional guidance using focus lines has the potential to achieve both a high number of obtained targets and attentional guidance effects compared to conventional arrows and lighting. It also suggests that it is possible to enhance the attentional guidance effects while minimizing interference with the immersion by adjusting the parameters of the focus lines.

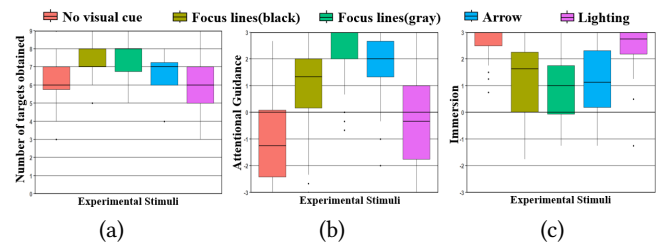


Figure 2: Results (a) Number of obtained targets (b) Attentional guidance (c) Immersion maintenance

4 Conclusion

This study proposed a method of applying “focus lines” to guide the attentional in VR space. The results of comparative experiments showed that focus lines (especially gray ones) had a high number of obtained targets and attentional guidance effects, suggesting the possibility of guiding the attention while maintaining the immersion.

References

- Sathaporn Hu, Joseph Malloch, and Derek Reilly. 2021. A Comparative Evaluation of Techniques for Locating Out-of-View Targets in Virtual Reality. In *Proceedings of the 47th Graphics Interface Conference*. Canadian Human-Computer Communications Society. doi:10.20380/GI2021.32
- Lori McCay-Peet, Mounia Lalmas, and Vidhya Navalpakkam. 2012. On Saliency, Affect and Focused Attention. In *Proceedings of the 30th ACM Conference on Human Factors in Computing Systems (CHI '12)*. ACM, New York, NY, USA, 541–550. doi:10.1145/2207676.2207751
- Jan Oliver Wallgrün, Mahda M. Bagher, Pejman Sajjadi, and Alexander Klippel. 2020. A Comparison of Visual Attention Guiding Approaches for 360° Image-Based VR Tours. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 527–535.